

A Packing of 97 Lines in $\mathbb{R}^{12}$

Limin Ge*

October 8, 2026

Abstract

We give 97 1-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 0.859219. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.857694, obtained by deleting 1 of the 98 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 97 subspaces of dimension 1 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.859219 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 1$. The cell $N = 97$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.857694, obtained by deleting 1 of the 98 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(12 - 1)/12 \cdot 97/96 = 0.926215$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n1N97.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 83 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
859219680714692637718912400000577388950012784864
denominator:
1000000000000058045298265821630617653356949188185

B Coordinates

97 blocks of 12 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m12n1N97.txt`.

-415593196374	-432815430029	386761169552	410610132588
-101723036003	246392068746	94944770698	-294047866641
-172704115561	-388853198825	-569681487122	294325732268
460478338466	223848642916	79774050047	-81425475224
-569362176962	270820055577	-425823266016	-165990246650
270266558150	510722333697	218971502122	-222245374451
-286709680085	326145903055	-386091262999	-27454911368
-31155334378	-54787947558	69927699528	-17513871524
193314637077	-539860658209	60285670763	-36837360821
49319583367	55119053602	-321601990468	-46108840525
221318363903	13965915801	-398378821824	-106999161952
76762767082	8128558513	-322375998829	-87369971372
-41531880328	177655989942	463639307795	45518205898
36322355610	169767162696	159208318681	-530022661044
20247063945	640912840173	1007141656	96185531712
-27384092239	69235085213	-82254467220	-529395634311
178539600218	-28582740812	-211853611940	624284702318
128548621710	-85289169173	-480953166901	118892519899
548264001134	-93953599362	-261469336338	8078767505
-421852375947	-51012355735	-24135541512	533171191898
-591625017192	116079587785	-122944755955	-452793448616
107018076697	24839792338	-211699491087	-110435570596
-225644613348	106070780904	-502028523450	444022358442
237599362440	-675780438755	166378379299	350506687271
-180260130364	-146956806143	27579002163	136463245621
-332524814417	247585295877	428996879947	-12085870291
249529825455	-6174048350	-209561494914	-89011479495
-44389695130	311411652515	294095094541	287864002918
82998166596	75474042530	-340535534244	-255472737100
-65583610770	128927382447	-251581170328	60092977469
-295399462763	456846642550	-66649661593	-82488115462
-277948393326	-231051630182	-143408085128	-371463240986
497251348148	79659759767	-240205095232	-329182896940
209156262586	-324729634839	-333708716319	-279382282096
410481450874	-341196377371	408884062306	-527361016977
396852151101	257993910667	323042510367	-230971241172
-170600895971	557875986125	-141226475853	-120074669398
67629919435	112142308294	-69731588623	-249234965167
241714160186	-442907149629	-432136753018	-455721396156
-584447717245	496507578834	378922403805	-194228363729
143915244708	215095337690	133910724014	17541175611
520878726616	-221836119739	-168058129970	122830129101
333103925030	-392226000619	562431910859	640192118747
81753384484	-68623804573	141848057970	230578646937
96227154113	-179736436045	345854155606	113362384507
-208149304153	-148939364813	65136160497	60691338871
183559028383	76662846704	-206853453141	-621780699737
265271290775	265788538119	-152280189237	96486484130
-129477914361	320783920961	-11133116537	-30837970898
17067811608	263390521269	-624255066624	49674082678
386192021930	28607833034	-92009802388	68079761627
-59063414031	423691208247	-171038870882	27792198839
-412388883857	-327452779694	-51744960935	297943916924
78372052342	-363554122876	-205533461517	-164121469713
351950123662	-51153041242	-612000165041	122605683750
436529007606	55804938056	78498600424	222654889935
496645967998	305103828521	-5327500774	268045727246
182292616765	461421986242	-32813854251	158654570755
142964070253	380224437659	134524195709	-152600307943
198372426607	-302712948733	-226365171848	426462939102
-132003520945	21681655083	165410723532	-246193189976
-185237750466	-178508156556	240209851493	-129592862017
212062006731	-365896360567	-634884915853	622813908028
54391155330	4978161053	-125653510859	-293118459940
-64733856029	303664279593	189741191673	-45403521649
139231815490	378356435938	-317417093577	282724065348
-371417143924	-157798002324	468603126075	160876795284
211749928096	-310610483417	124839904688	306505021876
-587391701418	-126234031852	-304562082775	-401773534488
-317649803188	62576062434	331729367188	365238358221

189333067320	503220536169	145432931292	-59166098076
-23311283266	-100133266067	-151093208953	-45834244761
-249049462198	253359787195	114308754398	-122585760817
275815101040	-393802317940	545579448308	20668428907
362909622832	244532168884	302842968643	133041218918
-274010057198	-279167411697	312715855998	12236029058
-445049938998	-384739423841	101365126701	423507242951
77510722426	418111122526	-92486383241	-505919595366
-347683193861	-403982515512	-483175705449	-335751110041
618835377368	120493805905	57819163550	395722776081
305083567283	-5952030503	141487990441	4459876867
-28634504015	9395500801	64845755706	-163801724030
253036327572	117832459629	-342059039986	-300848641558
-208608357316	358368485586	-179135436255	-380390214541
203189703058	-87457826600	-237031698899	465639972247
324366241418	-267110836720	688493488149	626232475363
-117564183578	46688725095	-213276861805	86963270424
-90328344652	627958173409	69659497112	-336178221499
-131670855331	127078764988	93667081991	124638409290
330001741683	-82901730907	-23667912569	37676678537
-24097674998	-19662617016	526043315170	-331150991
-217020980965	-294785479642	112453553738	-149928497024
218927884632	482850717806	40830227128	195719445766
217932303332	-176315529360	285417509860	80671669913
79378034136	221484534416	29119242303	228577445795
-129686272340	-317643569575	-214292159937	365977182996
-161951798704	218633899247	557339229611	34103061920
522081239997	-358333478912	4918510776	-235168685550
253132583602	215122352183	190442995422	-164889835080
-343575056216	-185407310178	308427537246	162381503814
-41889182429	-122902879853	-364312208724	-351300441153
592542026221	-417818410077	-77218081292	-128711530533
-199406567910	-94338271623	41231401927	590346157048
-327974121852	-226385009701	586728801222	197272328438
-169849292980	-142606289287	69154035456	-193654305186
25140318139	-59269262397	-431875144057	-150761696401
-41249760326	-674210866934	-35008780862	536125031984
256349718564	-122089814597	-404130501855	-122112719720
47267035460	-14310680253	-148246169360	105734626261
303701390123	337484909097	6480969014	-387038795465
-542751978252	172471950587	-202213268045	322704478328
240362026668	149768337767	-173197763520	340713609934
336136358058	-397212931173	200515480372	-65889823329
442676501795	-75564716176	-406313584256	315930378498
-631025708838	-52376020616	219655406493	-48149190825
72263527127	-55587603562	-91273433929	227092007992
181943933044	-792798763894	-657721953465	85042319644
80259211368	79352357989	29923341930	382218985146
-104805902446	171824169711	198047986335	529647462420
-533203770113	7100813459	155017876134	-164026322754
-134533417864	163197406850	-325794768719	-171802693101
-288395319200	-129873095247	-155772841385	380170890345
-277855638868	-190484653111	270045171543	-86212159598
-263133191789	79798852148	-113050856684	-233459978013
-31492330531	683322109917	309719529071	636594414471
-114206671382	-236427937758	366926497253	30310967424
-283332262584	-155617419928	-11676256468	-210485646704
256176483900	-82001937785	58953490608	-97138971261
329764742951	117117084249	40532430080	268571825892
-287059001580	561907855916	-520885874265	-445638495536
457071213060	-88743523870	327924204611	181395802548
248792911299	-150820051125	222765694059	-21851978050
-169353513440	309350188380	-367748335358	-359899909365
-39219968518	-228752396456	313754167586	-212624613357
-553447232367	-634421834105	156713244069	-153940116422
64104836278	50243778038	415814435894	-182419539816
-47982234195	-264961298808	361303139906	59003004070
-221816124708	-11408377375	-67699187272	34053144864
-617219675004	-170746918626	171283660573	351245587996
-267016971773	550983554309	-103357362524	-477726054763
344285972789	-19356550152	-53515464904	122739854663
299414182536	-51963390229	-385651555173	-445404258437
36773582343	38575225069	-214805545616	-179471028717
142097423430	-47237790450	257839930	-407911617789
284773763164	-144139448166	34453458304	-373390145835
2093243119	5608352274	-289027280309	100920703297
-14873325156	188004499042	359783029942	-276214235843
-259423583391	-131464155813	150899261299	-266290698939
330062948607	341174731199	-420000354892	-204866135373
245870519562	95232020971	590684453813	-19979764806
388739764871	322501311750	-13023312734	-467712888884
-66841540991	68322904951	49761822165	37223682526
-309771165259	91345242292	-22782273258	-281158568028
138591408652	637334268293	-10065813141	79260975047
198844045971	-2066188542	262832235622	-430855275851
-272510656655	527139647086	-270929786685	-411959689233
-29065264715	-227844993483	207093488061	184765126216
81382800440	-5551799076	85481548647	282349369015
-354793528911	-281405916779	-599944137159	-104909225379
114836204033	-676205385607	-614939909825	521776394983
-449803015931	218446357417	199720777426	105815109825
516323864202	-162709738271	161501737196	22703449714
-499171687869	-42598321210	-197429698987	-123680332752
11222554626	-451801876502	611623589343	-382745171978
382191114246	-29397206502	-92087728535	243190808662
-251180468238	141947175164	256407195729	410957395752
290792548298	230896922350	-146657898144	-94496220349
-454772052434	239067804887	201896980531	438661450713
184566714593	-206618017470	-187851566933	46642759320
271353314701	-125322357748	-582499476393	-119585513489
152141796007	-325375234749	-109739180538	157883511265
25869432487	522204219466	-241255933016	-587745052029

-536094605806	-314949016278	155820617376	186405083913
-329933241230	-273254157405	252062159753	-426281159120
-196353315647	397243156066	116909957428	-55644707717
162091810950	51484556828	-8388894712	-164466538614
95590740175	160051020566	-496078941524	-540813101058
331735650834	35565894884	422996112124	-153759646905
-6083415904	-267651281690	90641946381	-10805517222
-181238570055	-661225852302	-155245959305	514955210871
67170746249	121163307871	600729376141	168409266856
329544685899	448367949177	-355225938834	13555256847
-199520985103	-277866259215	318646468100	-224444924421
64609295509	-89566489923	290080694503	-382329615248
-679308471018	-49637854923	100708130164	-284442317052
-222054941797	-19975514207	-109827937241	287112590818
-338013820408	-477061320375	-339761890049	-548152486471
132152825678	235938308098	-3590094994	-176001559658
-16330549416	-327199605198	376476619125	225071017891
6478306902	519911834522	130971415563	-119080167514
-327744536920	-52960971606	-156542338671	-426126972735
304457158563	8436277215	78786888718	271396406306
-92167449872	-15736515968	-411967390207	27237851204
389847563412	95749424377	246075001900	-362257325505
73493761097	-375501157480	-181728048397	-145665149572
-81569090070	165897557439	167255071543	139983138044
9030193719	-356294196211	-48792822583	-393339044795
-462107720261	177812866461	51134026238	144822503841
-31596897277	-48668403207	486674024062	256157365462
-268923746541	119532827022	-228609024066	-156210690965
-264720464779	549316604075	521139383744	67550222426
578624960296	150195318791	-91741672592	-33186571120
-363718589674	-395986776284	-137960772416	-41923163087
57885858304	194014588893	339978725885	-31243262006
26027917243	-591479875155	-323769216925	112095295340
-177542779141	-121580422965	-317582077501	271141805275
366712312379	-66699660061	-524801677563	190959464615
465788271289	12566654035	331721041088	232524227486
495745381829	-112412102155	54521226915	-50711673980
-273479442069	-287602588097	-30916176211	-849938591249
276601043183	11513031383	191666213251	-178493850085
283901049450	523552806681	-478062926867	-391902148180
-319555874344	117517271282	-195498128863	-151485052150
150547067989	446047018860	129932923499	169553102026
-70450074410	-259252619166	286900929956	-71238673956
94649110288	-371777951664	47294484438	35421188024
-453453510202	2002183412	-356849191523	-463529727119
-102173608942	99188142006	137755531696	-445655340434
-546406395068	270686090543	-199571821089	178458544951
156387167652	-395474134136	73159646243	-409904445059
-84064471968	256225159682	-622651782904	-329309688586
-258236663998	48954179216	-291608115001	186399032593
-7061627124	459971516121	268097215947	58525290320
197728308276	560425727689	61973927401	18248652815
134512039063	266933489232	205791044396	284653728723
-548349139374	359636477905	-19727614037	35998781129
120272585611	268532047733	-130022372402	271241909386
121566649364	-168593795822	-447670025786	255763617087
-150624370555	-205126638276	352695107636	318249098335
243884696556	-47793865258	-163861976797	69244504365
379083071819	-271728663911	-60394046511	95715228177
304470214683	-67470396263	579206837361	-242019654286
-16503647733	-217503877579	-213425422909	-726962401347
701844936696	-56268519867	60203752079	269526209112
-6596097724	194465919098	-212100529651	238860843857
122852693372	332026021156	-3005847548	503182885658
-213377002218	-222341853291	-292127099952	-156944694479
182174900622	295704371628	-309264486052	-45780181526
-82095008432	-240350440255	333753548775	-86385361437
296604886311	165684655588	350831360380	413702485510
-2956599879	456056944165	354471979936	-428034504642
-247055400973	421038787981	371896960063	293125879097
-164598005208	-74314281539	-594241450569	-237930656713
313294872978	-75688391178	137980914297	-369490437869
161353241784	-38749183127	-204088145778	110221091883
528156006277	481487872882	-332556341864	99968260874
-617204063769	-173581765072	162622579224	-275654711695
255000318241	-184206603990	-258679292928	-134868968877
-7357262527	496362700153	-244801751527	-77193153169
60547398685	-394072165932	-110112219570	77060594057
-140218344118	225636608686	204919816471	46746147540
-198438281464	-453849952529	65031769127	367589994934
41241367433	-256824414586	177608690783	756019645096
163107658529	91162862560	-285662555887	259306261842
-262851743199	-244005571637	-230707710046	5792442745
-130709927423	129557506612	-579108167630	175225012572
-107323623036	-16501769622	248589636127	-202329268585
342815346647	355684150445	-65819543690	-214440664424
-538743243439	-85437283437	-448716143401	222972845319
-295101168328	5614175934	227149950552	91699753711
-339777412056	-104020114594	4112611053	-39163708777
463839961317	-61681527359	-323526650465	-381069959959
96397680619	551874176743	-224186618067	226089117598
198142628345	301724699580	155696487034	583245417133
-563276504460	60061501190	426138656233	-82971217374
246303084683	717486688030	-221600365371	-35691046445
109136825503	-175055836680	230991372359	86040533347
180748522222	-178422168263	197824487219	-271278400748
52691856039	-4644441457	63916879366	-307444601113
298135554288	39320947752	-370949543109	-469211518077
-77723536248	383428470655	242645470140	94767197736
230400608868	-2897068054	-400332443572	244095645741
68841340194	-261981663121	-304750108833	-87396216148
-498083344875	474814507758	-81490186913	-193820617658

114465365299	-22616018852	-208146229083	250509316874
-576744841145	-175191922514	-425838416580	74212806585
-21541406845	511032060730	-301989550054	360392568213
353811499860	51989379063	-235444451181	27215288422
493806816331	-38634582659	-1901894729	-374402009086
293291962737	110091155962	308741557480	64214657173
-143255755167	230355975713	86909877914	584780220553
253891440592	418358200416	-151688029758	119866982563
229997052708	-624707829757	499019439909	-183156102373
117937550233	14895935108	498030079952	-438315789070
674357595400	-333315381753	-53123873580	320930851950
169612757365	226653233779	677492392453	334494444487
-339375580711	-258780008853	-159651504622	-88141701057
-140046938723	-136149099071	274592345996	126097007124
51538709423	342258217968	-164592387773	168235771980
-112882167546	75405522513	43364416879	-381260836077
-84382597682	8780991373	-263169960284	-40717677547